Machine Learning Project Proposal

DTSC 691: Applied Data Science

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Project Title: Predicting Financial Access Profiles in Kenya Using FinAccess 2024

Project Objective and Scope

This project will build an end-to-end supervised machine learning system that predicts an adult respondent's current financial access profile in Kenya using the FinAccess 2024 public survey. The main analytic sample will use adults aged 18 years and above. The target will be constructed directly from C1_10 (current mobile money use) and C1_12 (current bank use) after removing responses coded as unknown. Respondents who are not currently using either service will be labeled Excluded, respondents currently using mobile money but not bank services will be labeled Mobile money only, and respondents currently using bank services will be labeled Banked, whether or not they also use mobile money. This target therefore represents current access status, not past-year behavior, lifetime use, or future financial outcomes.

The main goal is to understand which background, household, and digital-access factors are associated with each profile and to use those factors in a practical prediction tool. The project will focus on prediction and interpretation rather than causation. Predictions will not be used to infer long-term financial well-being, creditworthiness, fraud risk, or any other downstream outcome that the survey was not designed to measure. Any relationships found by the model will be interpreted as correlational patterns only.

Machine learning is appropriate because the FinAccess survey is high dimensional and the relationships between financial access, demographics, household structure, location, and digital behavior are likely to be nonlinear. At the same time, I will not train a model on all 3,816 raw fields. Instead, I will narrow the survey to a small, interpretable set of candidate predictors so the final workflow remains computationally efficient and appropriate for a master's-level applied data science project.

Data Strategy

The dataset for this project will be the FinAccess 2024 public household survey for Kenya. The materials available to me include the weighted public Stata file, the questionnaire, the survey manual, and the main report. The full public dataset contains 20,871 survey responses and 3,816 fields, covers all 47 counties, and includes information about banking, mobile money, savings, credit, financial health, digital access, demographics, and physical access to services.

For descriptive summaries, I will use the survey weight variable (indWeight) so the EDA reflects the survey design more accurately.

To manage high dimensionality, I will not carry the full 3,816-field matrix into modeling. I will start from documentation-guided candidate variables and shrink the feature space to roughly 10 to 20 interpretable predictors; my current working shortlist contains 11 variables: county, sex, age, household size, education, marital status, number of children in the household, livelihood, internet access, internet frequency, and financial health. Candidate fields with more than 40 percent missingness, near-zero variation, unclear questionnaire meaning, or weak practical relevance will be dropped. If a survey section creates a large battery of multiselect indicators, I will summarize it into a smaller set of interpretable counts or binary flags instead of carrying hundreds of dummy variables into the model.

I will also define leakage carefully. Any variable that directly reveals current financial access status will be excluded from the predictor set. This includes the target-defining columns C1_10 and C1_12 and other current-use fields such as current primary financial provider or current account-use indicators that would restate the class label.

Missing-data handling will be explicit rather than generic. Known nonresponse codes such as 98, 99, or other special survey values will first be converted to missing values. Numeric features such as age, household size, and number of children will use median imputation. Categorical fields will use either an explicit Missing/Unknown category or most-frequent imputation, depending on the variable definition and the amount of missingness. Rare categories may be collapsed into an Other category when needed to keep the encoded feature space stable.

Methodology & Analysis

The project will follow a standard supervised machine learning workflow: exploratory data analysis, feature screening, preprocessing, model comparison, evaluation, interpretation, and deployment. The main analysis will be conducted on the adult (18+) sample so the prediction task matches real-world adult financial access more closely.

EDA

The exploratory stage will focus on understanding both the target and the selected predictors before modeling. I will produce weighted target-distribution bar charts, stacked bar charts by sex, education, internet access, and financial health, and a county-level view of exclusion using either a county map or a ranked county bar chart. For numeric variables only, I will use simple correlation plots where appropriate. I will also produce a confusion matrix and feature-importance visualizations later in the workflow so the final results remain easy to interpret.

Outlier detection will focus on numeric variables such as age, household size, and number of children. I will inspect summary statistics, histograms, and boxplots, then compare suspicious values against the questionnaire coding ranges. Only values that appear implausible or miscoded will be capped or removed. Valid extreme survey responses will be retained because

they may represent real households, and tree-based models are generally robust to moderate numeric extremes.

Data Preparation and Cleaning

Because most selected predictors are categorical, I will use a consistent preprocessing pipeline. County, sex, education, marital status, livelihood, internet access, internet frequency, and financial health will be treated as categorical predictors and one-hot encoded after basic cleaning. Numeric predictors such as age, household size, and number of children will remain numeric and be median-imputed when needed. I am choosing one-hot encoding rather than arbitrary ordinal coding for most survey responses so I do not impose an unsupported numeric distance between categories.

This preprocessing stage will also document which categories are collapsed, which variables are dropped for missingness or sparsity, and which fields are removed to prevent leakage. That documentation will make the workflow reproducible and easier to justify in the final report.

Model Training

I will begin with a transparent model comparison rather than a single algorithm. The core comparison set will include multinomial logistic regression, class-weighted logistic regression, random forest, class-weighted random forest, and scikit-learn's GradientBoostingClassifier. This mix gives me a clear baseline, a flexible tree ensemble, and class-weighted alternatives that may improve performance on underrepresented classes.

The workflow will use a stratified 80/20 train-test split and StratifiedKFold cross-validation so the three target classes remain represented throughout model comparison. I will start with modest, interpretable baseline settings. If the top models are close, I will use a small RandomizedSearchCV on the leading candidate only instead of a large exhaustive grid search. This keeps model training efficient on a 20,000-row survey while still allowing targeted tuning.

Because the Excluded class is smaller than the Mobile money only class, class imbalance will be handled explicitly. I will compare class-weighted models to their standard versions and use stratified splitting and cross-validation throughout. I do not plan to use ordinary SMOTE in the main pipeline because most predictors are categorical survey variables and synthetic rows can create unrealistic combinations. If minority-class recall remains too weak, I may test a categorical-safe oversampling approach such as SMOTENC as a sensitivity check, not as the default workflow.

To keep training efficient, I will use one shared preprocessing pipeline across models, a deliberately small feature set, and a limited set of candidate algorithms rather than many redundant variants. This approach is appropriate for the size of the dataset and still rigorous enough for comparison and interpretation.

Model Evaluation

The main evaluation metric for this project will be macro F1 score because the classes are imbalanced and I want performance on all three groups to matter. I will also report accuracy, balanced accuracy, class-level precision and recall, and a confusion matrix. In addition, I will

track recall for the Excluded class specifically because that group is policy-relevant and easy for a model to miss.

For interpretation, I will use feature importance and, where appropriate, permutation importance or SHAP-style explanations so the final model can be explained in plain language. The interpretation section will be careful to describe which features are associated with the prediction task without making causal claims.

User-Interface Integration

I plan to build the application in Streamlit because it works well with Python machine learning workflows and supports a clean, interactive interface. The application will include the required course web pages and a specific project page that explains the problem, approach, and findings for a non-technical audience while also providing the interface for model interaction.

- Biographical homepage with academic background, professional interests, and a brief personal introduction.
- Resume page covering education, work experience, and at least one image or graphic.
- General projects page with links to other completed projects.
- Specific project page containing the deployed model, plain-language explanation of the problem, approach, and results, and the input form for predictions.

On the specific project page, a user will enter key background factors through dropdowns and numeric inputs, such as county, age, sex, education, household size, internet access, internet frequency, and financial health. After clicking a prediction button, the app will display the predicted financial access profile, class probabilities, and a short plain-language explanation. The final course submission will also include links to the relevant project page, notebook, and repository materials.

Presentation Plan

The final presentation video will walk through the project problem, the FinAccess dataset, the target construction logic, the feature-screening strategy, the preprocessing choices, the model comparison results, and the final Streamlit application. I will explicitly include a walkthrough of both the main code notebook and the UI notebook or UI implementation so the full workflow is clear from raw data to deployed interface.

The presentation will also explain limitations up front: the model predicts current financial access categories, not long-term outcomes; the Banked group includes some dual users; and the results are predictive rather than causal. This framing will help prevent the model from being overstated during the presentation.

AI Usage Statement

I expect to use generative AI tools such as ChatGPT and Codex in a limited and transparent way during the project. The main uses will be brainstorming feature ideas, comparing modeling approaches, improving the clarity of documentation, and helping me understand coding errors. I will not use AI as a substitute for the actual analysis, model selection, or interpretation of results, and any meaningful AI use will be cited according to course requirements.

Project Completion Plan

Week 1

Set up the project workspace, review the course handbook and ML project guidelines, confirm the topic, and organize the FinAccess files and documentation.

Week 2

Review the survey manual and questionnaire closely, finalize the target definition, and complete the revised proposal.

Week 3

Incorporate mentor feedback, confirm the adult analytic sample, and lock the initial feature shortlist.

Week 4

Clean the data, create weighted descriptive summaries, document missingness and leakage rules, and finish the main EDA figures.

Week 5

Train the baseline and comparison models, evaluate class-level performance, and interpret the initial results.

Week 6

Build the Streamlit app, connect it to the trained model, and add plain-language result explanations and visuals.

Week 7

Finalize the notebooks, UI, presentation video, and report; confirm all required web pages are present; and submit the final materials.

Resources

- FinAccess Kenya reports and datasets: https://finaccess.knbs.or.ke/reports-and-datasets
- FinAccess methodology page: https://finaccess.knbs.or.ke/methodology
- FinAccess main portal: https://finaccess.knbs.or.ke/
- KNBS FinAccess reports category: https://www.knbs.or.ke/reports_category/finaccess/

- DTSC 691 ML Project Guidelines: https://docs.google.com/document/d/1t6TCFBMVVdyeBf2CVFPk11uC38S9HLdlZdJJhSK6m5k/edit?tab=t.yabt891547x3
- Planned Python tools: pandas, scikit-learn, matplotlib, seaborn, joblib, Streamlit, and possibly SHAP.